XML Introduction (Detailed Guide for PDF)

What is XML?

XML (eXtensible Markup Language) is a markup language used to store and transport data. It is both human-readable and machine-readable.

History of XML

- Developed by W3C (World Wide Web Consortium)
- Released in 1998
- Designed to store and transport data

Why Use XML?

- Platform independent
- Self-descriptive data
- Easy data sharing between systems
- Widely used in web services and configuration files

Features of XML

- Extensible (you define your own tags)
- Structured data format
- Supports hierarchical data
- Unicode support

Basic XML Syntax

```
<note>
  <to>Tove</to>
  <from>Jani</from>
  <heading>Reminder</heading>
  <body>Don't forget me this weekend!</body>
</note>
```

Rules:

- Tags must be properly closed
- Tags are case-sensitive
- Must have a single root element
- Attribute values must be quoted

XML Elements

```
<name>John</name>
```

XML Attributes

```
<student id="1">John</student>
```

XML Tree Structure

- Root element
- Parent element
- Child element
- Sibling elements

Well-Formed XML

An XML document is well-formed if:

- Proper nesting of tags
- Closing tags are present
- Correct syntax

XML vs HTML

XML	HTML
Stores data	Displays data
Custom tags	Predefined tags

XML	HTML
Strict syntax	Flexible syntax

XML Namespaces

Used to avoid name conflicts.

```
<book xmlns:ns="http://example.com">
  <ns:title>XML Guide</ns:title>
</book>
```

DTD (Document Type Definition)

Defines structure of XML document.

```
<!DOCTYPE note [
  <!ELEMENT note (to, from, heading, body)>
]>
```

XML Schema (XSD)

More powerful than DTD, used to define structure and data types.

Parsing XML

XML can be read using:

- DOM Parser
- SAX Parser

XML in Web Development

- Used in APIs
- Configuration files
- Data storage

Example Project

```
<students>
  <student>
    <name>John</name>
    <age>20</age>
  </student>
</students>
```

Advantages of XML

- Easy data sharing
- Platform independent
- Flexible structure

Disadvantages of XML

- Larger file size
- Slower than JSON

Best Practices

- Use meaningful tag names
- Keep structure simple
- Validate XML documents